

Exploring Student Difficulties in Ray Tracing through Drawing-Based Tasks

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Abstract

Ray tracing is a fundamental topic in introductory computer graphics courses. Previous research suggested that students find ray tracing particularly challenging. Ray tracing requires spatial understanding, mathematical skills, and programming skills, and it makes use of other computer graphics concepts such as illumination models. To investigate students' conceptual understanding and identify common misconceptions, we conducted a drawing-based user study with 23 participants enrolled in an introductory computer graphics course. We found that the most common errors were misunderstandings of geometric and ray tracing concepts, as well as process-related mistakes. Based on our analysis, we propose an integrated instructional approach that incorporates spatial reasoning training, AR/VR-based visualization, and drawing-based assessment to reduce cognitive load and improve students' understanding of ray tracing concepts.

CCS Concepts

• **Applied computing** → **Education**; • **Computing methodologies** → **Computer graphics**.

Keywords

Computing Education, Computer Graphics, Ray Tracing, Spatial Skills, Spatial Reasoning, Drawing, Visual Representations

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1 Introduction

Ray tracing is a computer graphics technique used to simulate how light interacts with objects in a scene to produce realistic

images [Glassner 1989]. As a fundamental topic, it is widely taught in computer graphics courses [Balreira et al. 2017]. Building an effective ray tracing program requires precise execution of several key processes, including ray generation, intersection computation, shadow determination, and lighting modeling [Pharr et al. 2023; Zou 2024]. Although the general concept of ray tracing is relatively intuitive, the underlying mathematical principles and algorithmic mechanisms are complex and abstract [Verschoore de la Houssaije et al. 2022a].

Prior work has examined performance on ray-tracing assessments, but in-depth analyses of the specific challenges and misconceptions remain limited. Using two years of quiz and exam data, Liu et al. found that difficulty with ray tracing tasks is linked to students' spatial ability and to their integration of conceptual knowledge, programming skills, and mathematical reasoning into coherent problem-solving strategies [Liu et al. 2025]. In order to better understand students' misconceptions in ray tracing, we exploit previous research, which has shown that drawing supports learning [Chang et al. 2020; Fiorella and Zhang 2018] and elicits learners' ideas and cognitive processes [Hsieh and Tsai 2017; Selwyn et al. 2009].

Motivated by this need for more research and our own experiences teaching ray tracing, we employ drawing-based tasks to assess students' conceptual understanding and identify common errors and challenges. The findings will help educators recognize pedagogical challenges and explore effective solutions while also establishing a theoretical foundation and providing targeted analyses of learning needs to inform the design of more effective computer graphics educational tools. In this research, we aim to answer the following research question:

RQ: *What ray tracing concepts do students commonly struggle with, as revealed by drawing-based tasks?*

2 Related Work

2.1 Ray Tracing Educational Tools

To get a better understanding of student problems with ray tracing, and due to a lack of studies investigating this problem directly, we have analyzed ray tracing teaching tools, their capabilities, and the authors' motivations for their design. Researchers have developed and experimented with a variety of tools to support the learning

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of ray tracing. In 1999, Russell developed a web-based tool to help students grasp concepts through real-time visualization [Russell 1999]. More recently, Virtual Ray Tracer was developed to enable real-time visualization and exploration of the ray-tracing process [Verschoore de la Houssaije et al. 2022b]. Subsequently, Van et al. released Virtual Ray Tracer 2.0, adding advanced features such as area lights, soft shadows, and acceleration structures. Participants reported positive evaluations of the tool, citing strong educational potential [van Wezel et al. 2023]. In contrast to tools that emphasize ray-path visualization, Rayground is an online educational platform that supports flexible experimentation, code modification, and immediate feedback on rendering results [Vitsas et al. 2020]. It was integrated into classroom instruction, and most students reported that the platform helped them grasp complex concepts and that it improved their learning outcomes [Vasilakis et al. 2021].

Ray-tracing tools can also be combined with established pedagogical models. Pan et al. developed an online, interactive computer-graphics (CG) tutorial that supports collaborative learning of CG concepts and algorithms in a virtual network environment [Pan et al. 2005]. Schweitzer et al. developed an interactive system covering 32 CG topics—including ray tracing—to help students visualize and experiment with fundamental rendering principles. Instructors integrated the Think–Pair–Share (TPS) model with these tools to promote active learning [Schweitzer et al. 2011]. We searched the papers presenting ray tracing teaching tools for information about specific learning barriers and challenges, but could not find any such information. In general, researchers' motivations included making their tools more engaging, supporting experiential learning, and visualizing concepts more effectively than static images in slides or textbooks. Accordingly, further research is needed to provide educators with insights and to help researchers develop more effective tools to enhance ray tracing education.

2.2 Learning Challenges in Ray Tracing

Ray tracing involves inherent mathematical complexity, including vector algebra, geometric intersections, and transformations. Mathematics is crucial to computer graphics in general, and inadequate mathematical knowledge is likely to negatively impact students' performance in learning ray tracing [Gálvez et al. 2008; Hui et al. 2012; Suselo et al. 2017; Tian et al. 2011]. Spatial reasoning presents an additional challenge, since ray tracing requires students to visualize and interpret three-dimensional scenes. Many students may lack spatial reasoning skills, making them unable to sketch three-dimensional scenes or effectively represent 3D concepts to construct appropriate mental models [Liu et al. 2025, 2022]. There is a lack of directly relevant literature on the impact of spatial ability on learning ray tracing algorithms. However, prior research has shown that spatial reasoning ability is closely associated with success across STEM disciplines, including computer science [Bockmon et al. 2020; Margulieux 2020; Parkinson 2020]. In particular, success in introductory computer graphics is linked to students' mental-rotation and spatial-visualization abilities [Liu et al. 2022].

2.3 Why Drawing in Science Education

Professionals in science, technology, engineering, and mathematics (STEM) fields frequently use drawings to reason about and communicate visual–spatial information [Arcavi 2003; Goldschmidt 1994]. Drawing supports comprehension of complex, abstract phenomena by enabling STEM professionals to externalize their thinking, derive new inferences from their representations, and iteratively refine those representations to explore new ideas [Arcavi 2003; Fish and Scrivener 1990]. Drawing tasks have been used across STEM to reveal and improve student understanding: Chin and Teou used them to elicit conceptions of biological inheritance [Chin and Teou 2010], and Cooper et al. provided empirical evidence that sketching supports learning in chemistry classrooms [Cooper et al. 2017]. Drawing is prevalent in education and recognized as a core professional practice [Brew et al. 2013].

Additionally, evaluating the coherence of the mental models of the students is crucial because students can produce accurate statements even when their underlying models are incorrect or contain misinformation [Vosniadou and Brewer 1992]. However, when students are asked to draw, their diagrams are often inaccurate, incomplete, or structurally incoherent [Harle and Towns 2013; Nyachwaya et al. 2011]. Many misconceptions become apparent only when students externalize their mental models through drawing [Harle and Towns 2013; Vosniadou and Brewer 1992]. Depicting structural relationships requires students to externalize their mental models and to clearly represent these relationships in diagrams [Fish and Scrivener 1990]. Many studies used student-generated drawings both as a distinctive method to generate concepts that are not easily captured by other approaches and as a formative assessment to diagnose students' ideas and support learning [Chang et al. 2020]. In line with this literature, the present study uses drawing-based tasks for the same purposes.

3 Methodology

We conducted a user study at the University of Auckland to gain deeper insight into the challenges and misconceptions students face when using drawing tasks to comprehend and apply ray tracing concepts. This study aims to establish a targeted theoretical framework and practical foundation for developing more effective teaching strategies and tools.

The user study was performed in a 12-week third year introductory computer graphics course, covering geometry, OpenGL, light and color, illumination and shading, 3D modeling and transformations, texture mapping, ray tracing, spline curves and surfaces, and image processing, aligned with commonly taught topics in introductory computer graphics courses [Balreira et al. 2018]. All students had studied ray tracing theory in a lecture delivered one week prior to this user study. To reinforce the integration of theory and practice, weekly tutorial sessions were conducted. These sessions provided exercises and conceptual explanations to help students consolidate their understanding of the lecture material.

The user study took place during the tutorial session held one week after the ray tracing lecture. During the session, the tutor distributed four carefully designed drawing tasks to the students as a learning exercise. The tasks were voluntary and not graded, but students were told that they would receive feedback from the

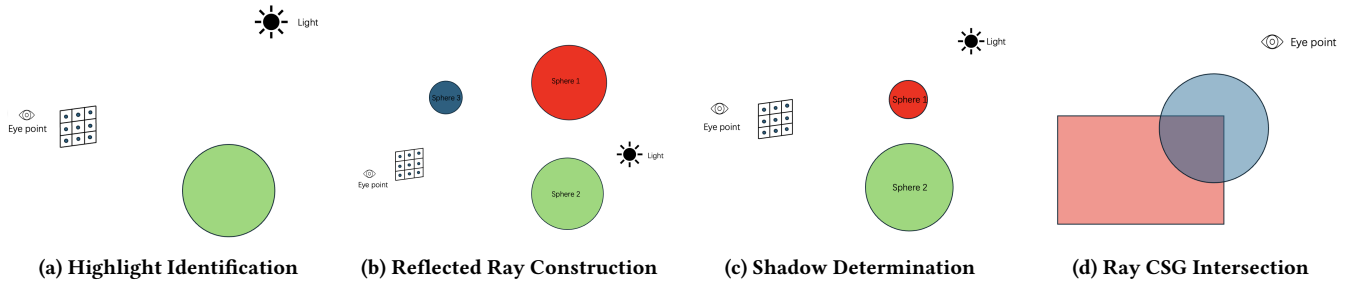


Figure 1: The four drawing tasks used in our user study.

instructors if they completed the task. Students were instructed to complete the drawing task independently within 10 minutes using paper and pen.

23 students completed and returned the drawing task. The students' drawings were scanned and photographed for record-keeping purposes.

3.1 Drawing Tasks

The drawing tasks addressed key ray tracing concepts, including highlight identification, reflected-ray construction, shadow determination, and ray - CSG Object intersection. The task details are outlined below and illustrated in Figure 1.

Task 1: Given an eye point, light source, and sphere (see figure 5 (a)) please indicate as precisely as possible where the viewer will see a highlight on the sphere. For this point, draw the view vector, light vector, and surface normal. Clearly label these three vectors, e.g., v (view vector), n (surface normal), and l (light vector).

Task 2: Given an eye point, light source, and three spheres (see figure 5 (b)), where Sphere 1 and Sphere 2 are reflective. Please identify one point on Sphere 1 and one point on Sphere 2 where the reflection of Sphere 3 can be seen. Indicate these two points as precisely as possible. For each of these points, draw the primary ray from the eye point to the point, the surface normal, and the reflection of the primary ray so that it intersects sphere 3. Clearly label these three vectors on Sphere 1 and Sphere 2, e.g., p (primary ray), n (surface normal), and r (reflected ray).

Task 3: Given an eye point, light source, and two spheres (see figure 5 (c)), please indicate the shadow cast by Sphere 1. Indicate in the image below which part of Sphere 2 is in the shadow of Sphere 1. Select one point on the surface of Sphere 2 that is in the shadow of Sphere 1 and mark it with a cross. For this point, draw the primary ray to the point, the surface normal, and the shadow ray. Clearly label these three vectors as p (primary ray), n (surface normal), and s (shadow ray).

Task 4: Given an eye point and a CSG object defined as the intersection of a rectangle and a circle (see figure 5 (d)), please draw four rays from the eye point toward the CSG object such that: 1. The ray intersects the rectangle but not the circle (ray R1) 2. The ray intersects the circle but not the rectangle (ray R2) 3. The ray intersects the circle and then the rectangle, but not both at the same time (ray R3) 4. The ray intersects both the circle and the rectangle, with part of the ray passing through both objects (ray R4) Please label the rays R1 to R4. Which of these rays intersects the CSG

object? For each ray intersecting the CSG object, clearly indicate the points where the ray enters and exits the object.

3.2 Data Analysis

We analyzed the user study data in three steps:

First, the four drawing tasks (Task 1 –Task 4) were evaluated using a pre-established scoring rubric, with task-specific criteria developed for each. The evaluation criteria are summarized as follows: Task 1 assesses view vectors, light vectors, surface normals, highlight identification, and geometric accuracy. Task 2 evaluates the ability to correctly locate the reflection point of visible Sphere 3 on Sphere 1 or 2, construct a complete light path, and ensure normal correctness and geometric accuracy. Task 3 assesses the primary ray, shadow ray, surface normal, and shadow region localization. Task 4 evaluates whether the four rays (R1–R4) satisfy the specified conditions and whether the CSG object is correctly identified.

Second, to characterize students' overall conceptual understanding, we categorized responses based on the number of correct indicators per task, as defined above: all indicators correct were labeled as "Understanding," at least one correct as "Partial Understanding," and none correct as "Not Understanding."

Finally, all errors identified in student drawings were included in the analysis. It is important to note that only submissions containing drawing results were considered; blank responses were excluded. Frequencies of each error type were calculated to identify common misconceptions with educational relevance. To ensure analytical reliability, two researchers independently performed the scoring and then compared their results. Discrepancies were resolved through discussion, and when necessary, adjudicated by a third researcher to finalize the results.

4 Result

4.1 Analysis of Student Performance on Highlight Identification (Task 1)

The performance of the students on Task 1, a ray tracing task that involves the identification of the highlights and the construction of the vector. Students performed best on highlight identification (74%), followed by surface normal (65%) and geometric accuracy of vector construction (65%). By contrast, fewer students (61%) correctly drew the light and view vectors. Because the specular reflection direction coincides with the view vector at the highlight, errors in these two vectors often co-occur. Notably, even when

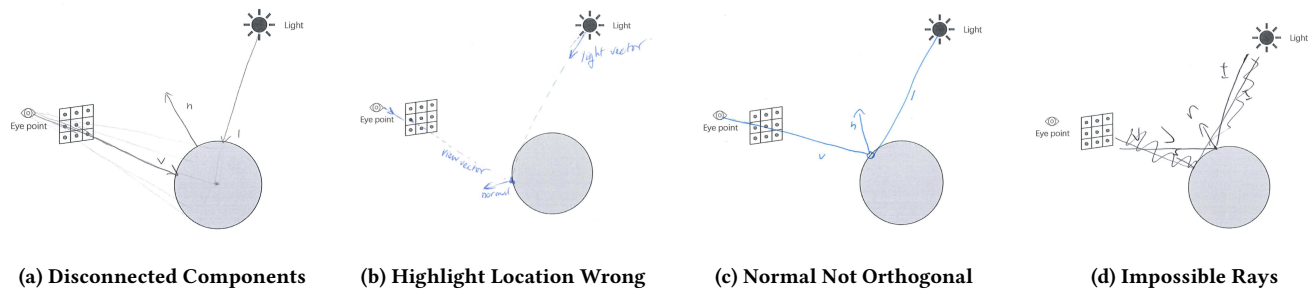


Figure 2: Four examples of student errors for drawing task 1 — Highlight Identification

students don't understand how to draw the light and view vectors, many are still able to find the correct highlight point.

Using these five criteria, we assessed students' conceptual understanding as shown in Figure 6a. Overall, 52% of students demonstrated full understanding, 43% showed partial understanding, and 4% of students did not understand how highlights are generated or how to answer the task.

We conducted an error-typology analysis of the 23 responses to task 1. The most common error was violating the law of reflection, i.e., the angle of incidence is equal to the angle of reflection (5 students), as shown in Figure 2b, 2c and 2d. The second most frequent error ($n = 4$) was that the surface normal was not orthogonal to the surface (Figures 2c and 2d). Additionally, 3 students misunderstood the mechanism of specular highlights and struggled identifying the highlight's location (Figure 2b). Furthermore, one student drew the view vector, light vector and surface normal without any connection, not intersection at the same point (Figure 2a), while another drew a view vector that did not start at the eye point (Figure 2d).

In several cases students' repeated attempts, as indicated by crossed-out lines, suggested that they understood basic properties, but were unable to find a solution, e.g., correct highlight location, where all basic properties (such as the law of reflection and normal orthogonal to surface) were fulfilled. This suggests difficulties of forming and manipulating a mental model of the problem, i.e., weak spatial reasoning skills.

4.2 Analysis of Student Performance on Reflected-Ray Construction (Task 2)

For task 2 students had to identify points on the reflective spheres where Sphere 3 is visible, and draw and annotate the required vectors. Based on the four evaluation criteria, the students performed best in geometric accuracy with an accuracy of 52%. Identifying a point on Sphere 1 where Sphere 3 is visible had the lowest accuracy, with an accuracy of 39%. 48% of students successfully constructed the ray path to Sphere 2, whereas 43% accurately constructed the surface normal.

Among the 23 user study participants, 30% exhibited no understanding, 39% showed partial understanding, and only 30% demonstrated full understanding, i.e., were able to: (i) select points on both spheres that satisfy the conditions for specular reflection, (ii) draw the surface normal through the sphere's center, and (iii) construct a reflected ray with equal angles of incidence and reflection such

that it intersects Sphere 3. The remaining 70% exhibited errors in point selection, normal orientation, or reflected-ray construction.

The three most common errors (observed seven times) were: (1) Failure to identify a valid intersection point on Sphere 1 or Sphere 2 (Figure 3c). This often resulted in missing the intersection point, the ray path, and the surface normal. (2) Drawing the primary ray to a point from which Sphere 3 was not visible (Figure 3d). (3) Lack of geometric accuracy, e.g., the law of reflection was violated (Figures 3a and 3d). The next most common issue (observed five times) was missing the surface normal at the intersection point (Figure 3d). Additionally, three students demonstrated a fundamental misunderstanding of the roles of the primary and reflected rays. In these cases, the primary ray was incorrectly drawn as originating from the object and reflecting back toward the eye point (Figure 3b). One student also misplaced the surface normal by drawing it away from the point of intersection, rather than at the actual point of contact with the sphere (Figure 3a). Overall, the core difficulty for students lies in identifying an intersection point on the reflective object with a line of sight to the target object, and using it to complete the full ray-tracing construction.

4.3 Analysis of Student Performance on Shadow Determination (Task 3)

For the shadow-determination task (Task 3) most students (78%) correctly identified shadow regions, indicating a high-level understanding that shadows arise from occlusion by Sphere 1. Accuracy was relatively low for surface normals (52%) and primary rays (43%), with the weakest performance observed in shadow rays (35%). This suggests that constructing shadow rays was more challenging than other components, revealing procedural gaps in translating conceptual understanding into accurate ray-path constructions.

As shown in Figure 6c, analysis of conceptual understanding in shadow determination revealed that only 35% of students demonstrated comprehensive understanding; they correctly identified that the shadow on Sphere 2 was caused by Sphere 1 blocking the light source and accurately labeled all required components. Another 43% of students demonstrated partial understanding; while most were able to identify the shadow regions and the basic mechanism of shadow formation, they failed to construct one or more of the required rays, indicating that procedural uncertainty persisted despite conceptual understanding. 22% of students exhibited substantial

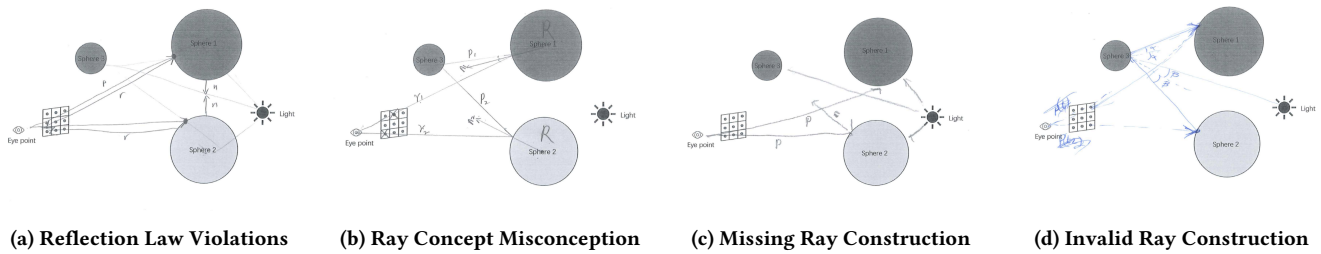


Figure 3: Four examples of student errors for drawing task 2 – Reflected Ray Construction.

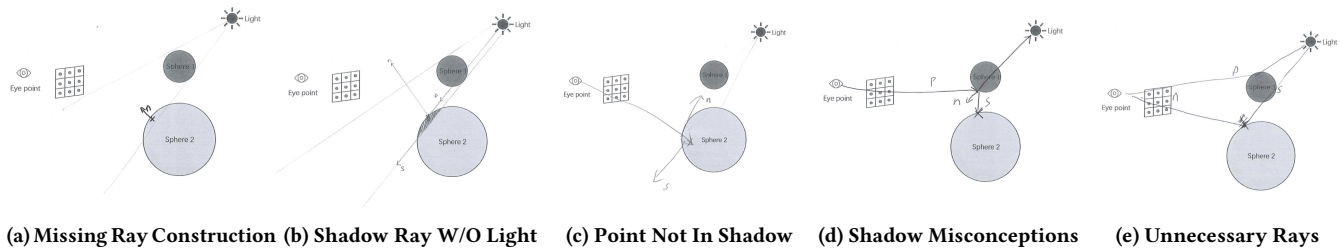


Figure 4: Five examples of student errors for drawing task 3 – Shadow Determination.

misconceptions, suggesting a lack of foundational understanding of shadow projection in ray tracing.

As shown in Figures 3b and 3c, the most common error (5 students) was the failure to construct a correct shadow ray: the ray did not proceed toward the light source to test for occlusion along the path. Additionally, three students incorrectly placed all rays on Sphere 1 rather than Sphere 2 (Figure 3d), indicating a misinterpretation of the task objective. Two students drew surface normals that were not perpendicular to the tangent at the marked points (Figure 3c). In two cases, more serious misconceptions were observed: in one, the surface normal was incorrectly drawn through the eye point and view plane (Figure 3e); in the other, the primary ray failed to pass through the eye point and view plane (Figure 3b). Finally, two students, as shown in Figure 3a, failed to draw any of the required elements (i.e., the primary ray and shadow ray), suggesting difficulties in visualizing and representing ray interactions.

4.4 Analysis of Student Performance on the Ray - CSG Object Intersection (Task 4)

Most students successfully constructed the four prescribed ray types. Specifically, R4 (intersecting both shapes simultaneously) had the highest completion rate (74%), followed by R1 (70%). R2 and R3 each reached (65%). These results indicate a solid procedural understanding of how rays intersect single and paired geometric objects and in what sequence. 61% of students correctly identified the CSG intersection region (logical AND), a rate slightly lower than their performance on ray construction. Compared with the other three tasks, Task 4 yielded the best overall performance.

61% of students demonstrated full understanding: they correctly identified which rays intersect the CSG intersection set and clearly labeled entry and exit points. 13% of students demonstrated partial understanding: they could draw rays that satisfied the conditions

or a subcondition, but were inconsistent in Boolean intersection judgments and in labeling entry and exit points. The remaining 26% showed no understanding, characterized by misclassifying multiple rays as intersecting the CSG intersection set and failing to identify the CSG object.

As illustrated in Figure 4a, seven students failed to identify the “ray-CSG intersection region” (despite generally correct ray constructions). In the example, the student identified all intersection points of rays with primitive objects, and the student does not seem to realize that only the darkly shaded area is the CSG object (formed by the intersection of a rectangle and circle). Two students omitted ray labels (Figure 4b), reflecting procedural and notation oversights. Few errors reflected genuine conceptual misunderstandings: one student misidentified all rays as intersecting the CSG object (Figure 4c), and another appeared to respond at random (Figure 4d), indicating minimal conceptual understanding. These patterns suggest that most errors did not stem from an inability to construct the required rays but from misreading the task and overlooking key steps (identifying the CSG intersection set, labeling entry/exit points, and labeling rays).

5 Discussion

This study investigates the conceptual challenges and common misconceptions that students encounter when learning ray tracing. Using a set of drawing tasks, it also examines the underlying cognitive mechanisms. The results reveal a pattern—“easy to identify, difficult to construct.” Students perform relatively consistently on procedural tasks. However, when they need to integrate spatial reasoning with ray-propagation principles into verifiable geometric constructions, both their conceptual understanding and task performance decline.

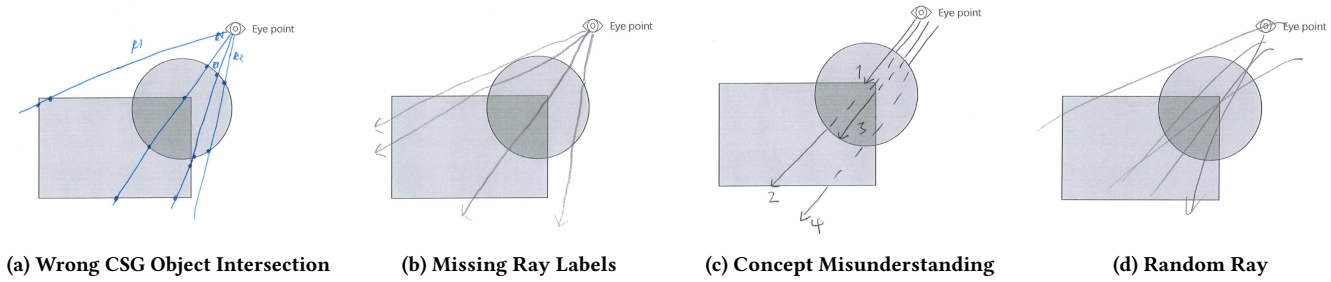


Figure 5: Four examples of student errors for drawing task 4 – Ray - CSG Object intersection.

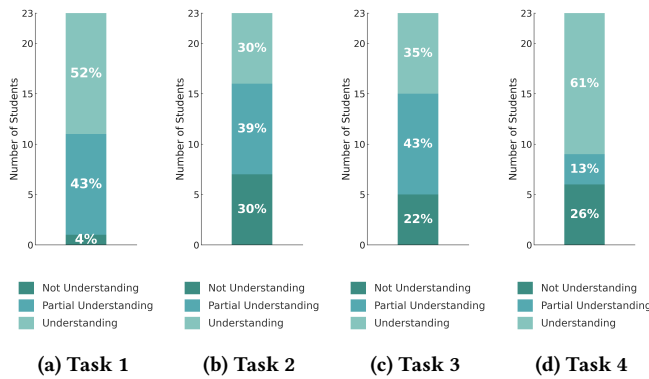


Figure 6: Student performance for the four drawing tasks.

5.1 Common Error

Based on the error types, we identified three categories. The first category comprises violations of core geometric and optical principles. Examples include violating the law of reflection (incidence angle $\neq$ reflection angle) with respect to the surface normal and drawing a normal that is not perpendicular to the tangent at the intersection of the ray and surface. These errors indicate limited quantitative geometric reasoning and weak precision control. A plausible contributing factor is students' limited understanding of geometry and linear algebra. In addition, some students' deviations may not reflect conceptual misunderstandings; instead, they often result from overlooked details or a reluctance to make precise adjustments. Second, misconceptions related to ray tracing concepts include initiating the primary ray from an incorrect position (not the eye point), failing to direct it through the virtual screen, misplacing the origin or direction of the shadow ray, and overlooking occlusion testing. Third, process-related errors involve students' inability to construct a coherent light path. Although some students were able to complete individual components—such as performing single intersection checks or adding local annotations—they struggled to form a coherent mental model that systematically integrates the eye point, ray, object, and light source. As a result, they were unable to construct a verifiable ray propagation sequence.

5.2 Learning Challenges and Possible Reason

Students demonstrated relatively strong performance on procedural tasks, particularly on the “Ray–CSG object intersection” (Task 4). The completion rate for each sub-task exceeded 60%, with an overall conceptual understanding rate of 61%, the highest among the four tasks. These tasks primarily rely on rule memorization and sequential execution and can often be completed by following a checklist. In contrast, both completion rates and conceptual understanding declined substantially on tasks involving spatial integration. The most challenging task involved the multi-sphere reflection (Task 2), with only about 30% of students completing it correctly. In Task 1 and Task 3, most students were able to identify the highlight (74%) and the shadow region (78%), respectively. However, accuracy declined when students attempted to translate conceptual understanding into precise ray-path constructions; only about 35% correctly drew the shadow rays. The main challenges appeared to lie in students' lack of a stable mental model and difficulty using spatial reasoning to construct accurate ray paths. Tasks that required students to visualize and construct three-dimensional relationships within a two-dimensional view while simultaneously maintaining multiple constraints. Students with lower visual spatial skills may have struggled to translate visual representations into appropriate actions [Gilhooly and Murphy 2005; Liu et al. 2022].

Additionally, students face challenges in vector precision control. Many struggled to consistently maintain key geometric relationships such as the equality of the angle of incidence and the angle of reflection, as well as the orthogonality of surface normals, revealing deficiencies in geometric reasoning and graphical accuracy. It is unclear whether this was due to weak mathematics knowledge or inability to construct a mental model. Furthermore, conceptual gaps also affect drawing decisions and path construction. Some students seemed to lack fundamental understandings, such as “the primary ray originates from the eye point” and “the shadow ray points toward the light source for occlusion testing,” which prevents them from completing the full light-path construction. Based on these findings, limitations in spatial ability and in integrating multiple constraints may constitute the primary bottlenecks for students when constructing correct ray paths. When combined with weaknesses in linear algebra foundations, imprecise vector control (e.g., reflection symmetry and normal direction), and gaps in core conceptual knowledge, these limitations collectively exacerbate errors in advanced tasks such as reflection and occlusion.

Our findings suggest that most students had a correct basic understanding of concepts, but were unable to integrate knowledge of basic concepts in order to form a consistent solution. This observation is consistent with Liu et al.'s conclusion that, in multi-object, multi-step reasoning scenarios, the integration of spatial reasoning with geometric construction constitutes the primary bottleneck [Liu et al. 2025]. Simultaneously conceptualizing three-dimensional relationships within a two-dimensional representation while maintaining global constraints—such as ray propagation, reflection, and visibility—imposes a substantial cognitive load on most students. The core challenge is students' inability to construct a stable and consistent mental model that clearly maps the primary rays from the viewpoint, the light source, and the scene geometry onto image formation on the virtual screen. Students often struggle to accurately determine the spatial positions and propagation paths of rays within the scene. Even when they understand the principles underlying the formation of highlights and shadows, they still find it difficult to construct verifiable ray paths. While students can follow procedural drawing rules, they struggle to integrate quantitative precision, spatial reasoning, and logical verification into a coherent problem-solving process. This incomplete mental model becomes particularly evident when reasoning must be externalized through diagrams [Harle and Towns 2013; Vosniadou and Brewer 1992].

5.3 Teaching Recommendation

Computer graphics involves a substantial amount of mathematical content, and existing literature has reported both correlations and causal relationships between spatial skills and mathematical achievement [Lowrie and Logan 2018]. Moreover, success in computer graphics is associated with students' abilities in mental rotation and spatial visualization, which are core components of spatial ability [Liu et al. 2022]. Mathematical and spatial abilities are therefore integral to the learning of computer graphics. Although some curricula have removed mathematics as a prerequisite, mathematical literacy remains indispensable. Existing evidence indicates that spatial reasoning can be enhanced through training [Lowrie and Logan 2018]. However, such training is frequently overlooked in university curricula. Building on this evidence, we concur with Sudelo et al. that the systematic integration of training in spatial skills is essential in computer graphics and related computing courses [Suselo et al. 2022].

As instructional technologies, AR and VR are particularly well suited to spatial-skills training and the visualization of concepts in computer graphics [Dünser et al. 2006; Pathak et al. 2020]. Their immersive and interactive features can reduce cognitive load and enhance comprehension of abstract ideas by providing intuitive three-dimensional (3D) representations [Oberdörfer et al. 2019; Suselo et al. 2018, 2021, 2022]. We believe that students may benefit from AR/VR visualization tools that help them construct appropriate mental models and enhance their learning.

Furthermore, this study is consistent with prior research on the use of drawing in science education [Ainsworth et al. 2011; Cooper et al. 2017] and further demonstrates the applicability of drawing tasks to computer graphics education. Such tasks function both as effective learning activities and as sensitive diagnostic instruments that elicit students' mental models and pinpoint bottlenecks where

“recognition is easy but construction is difficult.” For future practice, we recommend integrating drawing into formative assessment and targeted reinforcement, and aligning it with programming assignments. Specifically, students could first construct models through hand-drawn sketches, then engage in AR/VR-based learning activities, and finally perform code-based verification to promote transfer from concept to implementation.

6 Limitation

This study has several limitations. First, the sample is small ($n = 23$) and drawn from a single course at a single institution, which constrains external validity and limits the generalizability of the findings to broader contexts. In particular, the extent and emphasis of mathematics instruction vary across institutions and regions. At the study institution, mathematics is not a required prerequisite, which may be associated with comparatively weaker mathematical preparation among students. Second, the drawing-based tasks targeted a subset of ray-tracing concepts. Although these tasks effectively surfaced conceptual misunderstandings, they may not fully capture the challenges students face when implementing ray tracing in programming or applied rendering environments. Third, the analysis is based on static, paper-based drawings. While useful for identifying reasoning processes, these artifacts may not fully reflect students' dynamic problem-solving strategies or their iterative adjustments when using interactive visualization tools.

7 Conclusion

We conducted a user study using drawing-based tasks to identify common errors and challenges in students' solutions to ray tracing tasks. The results indicate that students performed relatively well on procedural tasks but struggled with tasks that required integrating spatial reasoning, geometric precision, and multiple constraints to construct coherent ray paths. The analysis revealed three major categories of common errors. First, many students failed to apply fundamental geometric and optical principles correctly. Second, substantial conceptual gaps were evident. Third, students struggled to construct accurate ray paths.

These issues may be caused by insufficient foundations in geometry and linear algebra, compounded by limited spatial reasoning abilities. Students may lack the ability to construct stable mental models of the ray tracing process. They often fail to produce verifiable geometric constructions that accurately represent these concepts. This disconnect indicates cognitive overload when students need to process complex three-dimensional spatial relationships within a two-dimensional representational space.

To address these challenges, we recommend explicitly integrating spatial reasoning training into computer graphics course. As immersive, interactive tools, augmented and virtual reality (AR/VR) hold strong potential to enhance conceptual visualization, reduce cognitive load, and support the development of accurate mental models. Additionally, drawing-based tasks can serve as effective pedagogical tools for both facilitating learning and revealing latent misconceptions.

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